Two-Level Quantum Systems

(The simplest non-trivial quantum system)

A two-level quantum system is the simplest system that can exhibit all fundamental quantum phenomena: superposition, interference, relative phases, and non-trivial time evolution.

It is the minimal Hilbert space in which quantum mechanics becomes genuinely different from classical physics.

Hilbert Space of a Two-Level System

A two-level system is described by a Hilbert space of dimension 2:

$$H \cong C^2$$

We choose an orthonormal basis, often written as:

$$|0\rangle \qquad , \qquad |1\rangle$$

These basis states may represent:

- two energy levels of an atom (ground/excited)
- spin-up & spin-down of a spin $\frac{1}{2}$ particle
- two stable states of a superconducting qubit
- any physical system with exactly two distinguishable states

A general pure state is a linear combination:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

with complex coefficients satisfiying the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

As established earlier:

- the **global phase** of $|\psi\rangle$ is physically irrelevant
- the **relative phase** btween α & β carries physical information

Thus, the physical content of a two-level state is encoded in:

- the probability amplitudes $|\alpha|$, $|\beta|$
- the relative phase between the two components

Energy Eigenbasis

Often, the two basis states correspond to two energy eigenstates:

$$H|0\rangle = E_0|0\rangle, \qquad H|1\rangle = E_1|1\rangle$$

In this basis, the Hamiltonian is diagonal:

$$\begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}$$

If the system is prepared in an energy eigenstate, its time evolution is simple:

$$|0(t)\rangle = e^{-iE_0 t/\hbar}|0\rangle, \qquad |1(t)\rangle = e^{-iE_1 t/\hbar}|1\rangle$$

Each eigenstate acquires only a phase factor.

No transitions occur between the two levels.

Superpositions and Relative Phase Evolution

If the initial state is a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Then at time t:

$$|\psi(t)\rangle = \alpha e^{-iE_0 t/\hbar}|0\rangle + \beta e^{-iE_1 t/\hbar}|1\rangle$$

The **relative phase** evolves as:

$$\Delta\phi(t) = \frac{(E_1 - E_0)t}{\hbar}$$

This time-dependent phase difference leads to interference effects when measuring in any basis different from $\{|0\rangle, |1\rangle\}$.

Thus, even without transitions between levels, a two-level system already exhibits non-trivial dynamics.

General Hamiltonian for a Two-Level System

The most general Hamiltonian acting on a two-dimensional Hilbert space is a 2×2 **Hermitian matrix:**

$$H = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}; \qquad a, b \in R, c \in C.$$

- The diagonal terms a and b correspond to the energies of $|0\rangle$ and $|1\rangle$
- The *off-diagonal* term c represents a **coupling** between the two states

If $c = 0$, the system has no transitions between levels

If $c \neq 0$, the Hamiltonian mixes the two states, and the system can oscillate between them.

This is the simplest example of quantum transitions.

Coupled Two-level system: basic dynamics.

Consider a Hamiltonian with real coupling:

$$H = \begin{pmatrix} E_0 & V \\ V & E_1 \end{pmatrix}$$

This Hamiltonian is no longer diagonal in the $|0\rangle$, $|1\rangle$ basis.

The true energy eigenstates are linear combinations of $|0\rangle$ and $|1\rangle$.

Diagonalizing H gives two eigenvalues:

$$E_{\pm} = \frac{E_0 + E_1}{2} \pm \sqrt{\left(\frac{E_1 - E_0}{2}\right)^2 + V^2}$$

The corresponding eigenstates are superpositions of $|0\rangle$ and $|1\rangle$.

If the system starts in $|0\rangle$, the time evolution leads to **oscillations** between $|0\rangle$ and $|1\rangle$.

this phenomenon is the simplest form of coherent population transfer.

The oscillation frequency is determined by the energy splitting:

$$\Omega = \frac{E_+ - E_-}{\hbar}$$

This is the origin of **Rabi oscillations**, which will be studied in more detail later.

Why Two-Level Systems Matter

Two-level systems are fundamental because:

- they are mathematically simple
- they capture all essential quantum behavior
- they form the basis of quantum information theory
- any qubit is a two-level system
- any finite-dimensional system can be decomposed into two-level subsystems

In quantum computing, the states $|0\rangle$ and $|1\rangle$ become the logical states of a **qubit**.